SmartECG Assist

■■■■■ ■■■■■■ ■■■ — ECG Heartbeat Classification

Afeka · Machine Learning · 1D CNN · MIT-BIH

■■■■ 2026

SmartECG Assist — [redacted]

[redacted]: ECG Heartbeat Classification

[redacted]: Afeka — Machine Learning

[redacted]: 1D CNN [redacted] ECG

[redacted]: MIT-BIH Arrhythmia Database (PhysioNet)

[redacted]: [redacted] 2026

[redacted]

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1. [redacted]

1.1 [redacted]

[redacted] Machine Learning [redacted]. [redacted] ECG ([redacted]"[redacted]), [redacted]:

[redacted]	[redacted]	[redacted]
Normal	[redacted]	0
Abnormal	[redacted] ([redacted])	1

1.2 [redacted]

[redacted]:
- **Pre-training** [redacted]
- **Fine-tuning** [redacted]
- [redacted] (`individuals`)

[redacted] (SmartECG Assist):
- [redacted] **1D CNN** [redacted]
- [redacted] fine-tuning per-patient
- [redacted] ([redacted] data leakage)
- [redacted] ECG [redacted]

1.3 [redacted]

```
MIT-BIH CSV + Annotations ↓ Preprocessing ([redacted], [redacted], [redacted], resample) ↓
[redacted]: Train / Val / Test ([redacted]) ↓ [redacted] 1D CNN ↓ [redacted] best_model.pth ↓
[redacted] Test ↓ [redacted] (SmartECG Assist) — [redacted] CSV [redacted]
```

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2. [redacted]

#	[redacted]	[redacted]
1	[redacted] — [redacted] / [redacted]	[redacted] 0/1, BCELoss, Sigmoid
2	Preprocessing: [redacted], [redacted], [redacted]	`data_processing.py`
3	[redacted] MIT-BIH	`dataset/mitbih_database/`
4	[redacted] Train / Validation / Test	`split_patients_by_id()`
5	1D CNN	`models/cnn1D.py`

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3. [redacted] MIT-BIH

3.1 [redacted]

- [redacted]: MIT-BIH Arrhythmia Database
- [redacted]: https://www.physionet.org/content/mitdb/1.0.0/
- [redacted]: `dataset/mitbih_database/`

3.2 [redacted]

[redacted] (record) [redacted]:

[redacted]	[redacted]
`{id}.csv`	[redacted] ECG — [redacted]: sample #, MLII, V5 ([redacted])
`{id}annotations.txt`	[redacted] beat-by-beat

[redacted] (`100.csv`):

```
'sample #','MLII','V5' 0,995,1011 1,995,1011 ...
```

3.3 [redacted]

[redacted]	[redacted]
[redacted]	360 Hz
[redacted]	48
[redacted]	29
[redacted]	~30 [redacted]

3.4 [redacted] (Annotations)

Normal (0):
- `N` — Normal beat

Abnormal (1):
- `L`, `R`, `V`, `A`, `a`, `j`, `S`, `F`, `E`, `e`, `r`, `!`, `f`, `/`

[redacted]:
- `Q`, `?` — [redacted]

3.5 [redacted]

[redacted] **-0.9%** [redacted] abnormal [redacted] ([redacted]).

19 [redacted]:
101, 103, 107, 109, 111, 112, 113, 115, 117, 118, 121, 122, 123, 124, 207, 214, 230, 232, 234

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4. [redacted] (Preprocessing)

[redacted]-`src/data_processing.py` ([redacted]) [redacted]-`src/inference.py` ([redacted]).

4.1 [redacted] 1 — [redacted]

- [redacted] ECG [redacted] (Lead MLII)
- [redacted] annotations [redacted] ([redacted])

4.2 [redacted] 2 — [redacted] (Bandpass Filter)

[redacted]	[redacted]
[redacted]	Butterworth Bandpass
[redacted]	0.4 Hz
[redacted]	30 Hz
[redacted]	360 Hz
[redacted] (order)	3

[redacted]: [redacted] baseline wander [redacted].

4.3 [redacted] 3 — Normalization

- **Min-Max normalization** [redacted] [0, 1] [redacted]
- [redacted]: `(signal - min) / (max - min)`

4.4 [redacted] 4 — Segmentation ([redacted])

- [redacted] = **360 [redacted]** ([redacted]-360 Hz)
- [redacted] — [redacted] annotations:
- [redacted] beat abnormal [redacted] → **Abnormal (1)**
- [redacted], [redacted] Normal → **Normal (0)**
- beats [redacted] → [redacted]

4.5 [redacted] 5 — Resampling

- [redacted]: **128 [redacted]**
- [redacted]-CNN [redacted]

4.6 [redacted] 6 — Reshape

- [redacted]-`(N, 128)` [redacted]-`(N, 1, 128)` — [redacted], [redacted] 128

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5. [redacted]

5.1 [redacted]

[redacted] (Patient ID) — [redacted] beat.

> [redacted]-beats [redacted]-splits.
> [redacted] **data leakage** [redacted].

5.2 [redacted]

Split	[redacted]	[redacted]
Train	~65.5%	19
Validation	~17.2%	5
Test	~17.2%	5

- **Seed:** 42 ([redacted])
- **[redacted]:** `val_ratio=0.15`, `test_ratio=0.15`

5.3 [redacted]

Train (19):
100, 105, 106, 108, 114, 116, 201, 202, 205, 208, 209, 212, 215, 219, 221, 222, 223, 228, 233

Validation (5):
102, 104, 200, 213, 217

Test (5):
119, 203, 210, 220, 231

[redacted]: `output/general_model/patient_splits.json`

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6. [redacted] — 1D CNN

[redacted]-`src/models/cnn1D.py` — [redacted] CNN [redacted] ResNet (skip connections).

6.1 [redacted]

[redacted]	[redacted]
Input	`(batch, 1, 128)` — [redacted] ECG [redacted]
Output	[redacted] 0 [redacted]-1 (Sigmoid)
Threshold	≥ 0.5 → Abnormal, < 0.5 → Normal

6.2 [redacted]

```
Input (1, 128) ↓ Conv1d: 1 → 32 channels, kernel=1 ↓ ReLU ↓ 4 × Block: Conv1d → Conv1d → Skip Connection → ReLU → MaxPool1d (32 channels, kernel=5) ↓ Flatten → 160 features ↓ Linear: 160 → 160 ↓ Linear: 160 → 1 ↓ Sigmoid → [redacted]
```

6.3 Block ([redacted])

[redacted] Block [redacted]:

1. **Conv1d** — [redacted] + ReLU
2. **Conv1d** — [redacted]
3. **Skip connection** — [redacted] (residual)

4. **MaxPool1d** — ■■■■■ ■■■■■

6.4 ■■■■■■■

■■■■■	■■■ ■■■■■ ■■■■
`num_blocks`	4
`block_channels`	32
`kernel_size`	5
`num_features`	160

6.5 ■■■■■■■■ Loss

- **Binary Cross Entropy (BCELoss)**
- Optimizer: **Adam**, lr=0.001
- LR Scheduler: **ReduceLROnPlateau**

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7. ■■■■■ ■■■■■■

7.1 ■■■■■■ ■■■■■

■■■■■	■■■
Epochs	30
Batch size	32
Learning rate	0.001
Weight decay	1e-4
Weighted sampling	True

Device	CPU / CUDA

7.2 ■■■■■ ■■■■■

- **■■■■ `best_model.pth` ■■■ validation loss ■■■■■ ■■■■■**
- **■■■■■ ■■■■■**: **Epoch 5** (val loss = 0.4714)
- **■■■■ ■■■■■■** — **■■■■■** best model **■■■■■■ ■■ ■■■■■■ Test**

7.3 ■■■■■■

■■■■■:

```
python src/main.py --mode general-training --output_path ./output/general_model/
```

■■■■■:

```
python src/main.py --mode test --model_path ./output/general_model/best_model.pth
```

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8. ■■■■■■ ■■■■■

8.1 ■■■■■■■ ■■ Test Set (■■■■■■■ ■■■■■)

□□□□□□	□□□
Accuracy	84.07%
Precision	0.639
Recall	0.933
F1-score	0.758
Balanced Accuracy	86.98%

8.2 Confusion Matrix

```
■■■■■ Normal ■■■■■ Abnormal ■■■■■ Normal 5,064 (TN) 1,211 (FP) ■■■■■ Abnormal
155 (FN) 2,144 (TP)
```

8.3 ■■■■■■■

- **Recall ■■■■ (93%)** — ■■■■■ ■■■■ ■■ ■■■ ■■■■■■■■ ■■■■■■■■
- **Precision ■■■■■■ (64%)** — ■■ false positives (■■■■■■■ ■■■■■■■ ■■■■■■■■)
- **Overfitting** — Train accuracy ~99% ■■■■■ Test ~84%

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9. ■■ ■■■■■■■■■ — SmartECG Assist

9.1 ■■■■

■■■■ ■■■ ■■■■■■ ■■■ ECG ■■■ ■■■ ■■■■ **■-annotations** — ■■■■■■ ■■■■ ■■■■ CSV ■■■■■■:
- **■■■■■**: **■■■■ / ■■ ■■■■**
- **■■■■ ■■■■■■** (**■■■■■■**)
- **■■■ ■■■ ECG ■■ ■■■■■ ■■■■■■** abnormal
- **■■■■■■■■■■■** (**■■■■■■ ■■■■■■**, **■■■■■ ■■■■■■**)

9.2 ■■■■■

■■■■	■■■■■
`web/app.py`	■■■ Flask
`web/templates/index.html`	■■■■ ■■■■■
`src/inference.py`	■■■■■, preprocessing, ■■■■■, ■■■
`SmartECG_Assist.command`	■■■■■ ■■■■■■ ■■■■■ (Mac)

9.3 ■■■■■

■■■ 1 — ■■■■■ ■■■■■:

```
SmartECG_Assist.command
```

[redacted] 2 — [redacted]:

```
cd ECG-Heartbeat-Classification python web/app.py
```

[redacted]: **http://127.0.0.1:8080**

> **[redacted]:** [redacted] `index.html` [redacted] — [redacted].

9.4 [redacted]

[redacted]	[redacted]
MIT-BIH (`sample #, MLII, V5`)	[redacted]
[redacted] ECG	[redacted]
CSV [redacted] `;` [redacted] TAB	[redacted]
[redacted] `.txt`, `.tsv`	[redacted]
[redacted] ECG	[redacted]

9.5 [redacted]

- [redacted]: **360 Hz** (MIT-BIH)
- [redacted] "[redacted] (Hz)" [redacted]
- [redacted]-360 Hz [redacted]

9.6 [redacted]

1. [redacted] (360 [redacted])
2. [redacted] abnormal [redacted] ([redacted] 300 [redacted])
3. **[redacted]:** [redacted] ≥ 0.5 → Abnormal
4. **[redacted]:** [redacted] Abnormal → [redacted]; [redacted] Normal → 1 – [redacted]
5. **[redacted]:** 15 [redacted], [redacted] = [redacted] abnormal

9.7 [redacted]

- [redacted] **300** [redacted] (~5 [redacted]) [redacted]

- [redacted] MIT-BIH — [redacted]
- [redacted] — [redacted]/[redacted]

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10. [redacted]

```
ECG-Heartbeat-Classification/ [redacted] dataset/mitbih_database/ # [redacted] MIT-BIH [redacted] src/ [redacted] main.py # [redacted] – [redacted] / test [redacted] data_processing.py # preprocessing + [redacted] inference.py # [redacted] running.py # [redacted] models/cnn1D.py # [redacted] 1D CNN [redacted] training/ # train / val / test epochs [redacted] web/ [redacted] app.py # [redacted] Flask [redacted] templates/index.html # SmartECG Assist UI [redacted] output/general_model/ [redacted] best_model.pth # [redacted] patient_splits.json # [redacted] *.log # [redacted] docs/ # [redacted] SmartECG_Assist.command # [redacted] run_web.sh
```

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11. [redacted]

[redacted]	[redacted]
[redacted]	`python src/main.py --mode general-training`
[redacted]	`python src/main.py --mode test --model_path ./output/general_model/best_model.pth`
[redacted]	`python web/app.py` → http://127.0.0.1:8080
[redacted] (Mac)	[redacted] `SmartECG_Assist.command`

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12. [redacted]

1. [redacted] **ECG** [redacted] MIT-BIH [redacted] **1D CNN**.
2. [redacted].

3. [redacted] **84% accuracy [redacted]-93% recall [redacted]** test.
4. **SmartECG Assist** [redacted] ECG [redacted].
5. [redacted]-precision [redacted] overfitting.

[redacted] *ECG Heartbeat Classification — Afeka, Machine Learning.*